Georgia Institute of Technology
School of Electrical and Computer Engineering
Fall 2026

ECE 8803-AI4: AI and Machine Learning for Semiconductor Process Technology and Digital Twins

Instructor:	Prof. Asif Khan	**Credit Hours:**	3 (3-0-0-3)
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Office:	Clough UG Learning Commons 423	**Schedule Type:**	Lecture
Office Hours:	TBD	**Campus:**	Georgia Tech-Atlanta
Lecture:	TR 12:30 – 1:45 PM	**Grade Mode:**	Letter Grade
Location:	Allen Sustainable Education Building, Room 110		

Prerequisites: Graduate standing or permission of instructor. No prior machine learning experience required. Familiarity with semiconductor device physics or fabrication processes is expected.

1. Course Overview

This course introduces graduate students to the theory and practice of applying artificial intelligence and machine learning to semiconductor process technology, metrology, manufacturing, and digital twins. The semiconductor industry is undergoing a transformation driven by the increasing complexity of advanced process nodes and the explosion of data generated in modern fabrication facilities. AI/ML techniques and digital twin frameworks are now critical tools for process modeling, equipment monitoring, defect inspection, yield optimization, and accelerating technology development cycles.

Designed for students with a background in semiconductor devices and processes but no prior AI/ML experience, the course begins with a rigorous introduction to machine learning fundamentals using Python, then progressively applies these techniques to real-world semiconductor challenges. Topics include surrogate modeling for TCAD, physics-informed neural networks, virtual metrology, SEM defect classification, yield prediction, fault detection, digital twins for fab modules, and ML-assisted technology development.

The course emphasizes hands-on learning through four Python labs using open semiconductor datasets, two problem-set homeworks, and a substantial semester-long research project. Industry guest lectures provide direct exposure to how these methods are deployed in production fabs and R&D environments.

2. Course Outcomes

After successfully completing this course, students should be able to:

1. Apply machine learning techniques (regression, classification, deep learning) to semiconductor process modeling and optimization problems.
2. Build and evaluate virtual metrology models that predict wafer-level measurements from equipment sensor data.

3. Train and deploy convolutional neural networks for semiconductor defect detection and classification on SEM and wafer-map imagery.
4. Implement physics-informed machine learning approaches that integrate domain knowledge of process physics into data-driven models.
5. Analyze large-scale fab datasets (sensor traces, yield maps, metrology records) using statistical and ML methods in Python.
6. Assess the computational and practical trade-offs of deploying AI/ML solutions in semiconductor manufacturing environments.
7. Critically evaluate published research on AI/ML applications in semiconductor process technology and metrology.
8. Design, execute, and present an original semester project applying AI/ML to a semiconductor process or metrology challenge.
9. Design and deploy an end-to-end AI/ML solution following the CRISP-DM framework, including data ingestion, modeling, evaluation, and deployment of an interactive demo via Jupyter notebooks and Gradio.
10. Communicate AI/ML results to technical and business audiences, including quantified manufacturing impact (yield, cost, cycle time), reliability considerations, and corner-case analysis.

3. Course Modules

The course comprises fourteen modules. Modules are listed in logical rather than calendar order; the session-by-session sequence is given in the weekly schedule below.

Module 1: Introduction — Where AI Enters the Fab

The course opens with the three exponentials that drive the industry — logic, DRAM, and NAND — the process flow that realizes them, and the six places where AI has already entered the fab: defect detection, virtual metrology, digital twins, computational lithography, predictive maintenance, and factory-scale orchestration. It then asks whether any of this is new, tracing shipped machine learning in semiconductor manufacturing back to the neural process models of 1991–1993 and arguing that the historical bottleneck was never the algorithm but the absence of a common data plane.

Module 2: Fault Detection and Classification

Every process run is a bundle of sensor traces living in a space of thousands of dimensions, and this module shows how principal component analysis compresses a run into a handful of numbers — eigentraces — that turn out to correspond to the physical knobs that actually varied. Building on that representation, it develops the six steps of a production FDC system: trace alignment, Hotelling's T^2 and squared prediction error as complementary detectors, contribution plots for diagnosis, ROC-based threshold selection, and the alarm-rate arithmetic that determines whether a deployed system stays switched on.

Module 3: Toolchain, Workflow, and Deployment

The distance between a notebook that produces a good number and a model a fab will actually run is the subject of this module: the CRISP-DM project framework, the working stack of Jupyter and Colab, GitHub, PyTorch, MLflow for experiment tracking and Gradio for interactive demonstration, and the reality that fab data arrives from historians, MES, and SQL databases owned by different teams. It then covers the production side — deployment, monitoring, drift detection and event-driven retraining, and interpretability with SHAP together with its documented limits on correlated inputs — and sets the deliverable standard for the semester project.

Module 4: Defect Classification

Using the WM-811K wafer map dataset, this module builds the same classification problem four times — logistic regression on flattened pixels, a deep network, a convolutional network, and hand-crafted polar features fed to a random forest — and shows that the physics-derived features beat the CNN precisely where the physics holds, and lose where it does not. Along the way it establishes the machine learning fundamentals the rest of the course depends on (gradient descent, backpropagation, softmax, macro-F1) and the two traps that decide real results: leakage from splitting by wafer instead of by lot, and severe class imbalance that renders aggregate accuracy meaningless.

Module 5: Statistical and Advanced Process Control

A century-old discipline, presented as the foundation on which every fab monitoring system is still built: the separation of common from special cause, control charts and the Western Electric rules, subgroup design, process capability, and the measurement system analysis that determines whether a reported Cpk is real or fictional. The module closes by moving from detection to compensation — feedforward and run-to-run control, EWMA controllers and their failure mode of hiding the fault they correct — and from univariate to multivariate monitoring, arriving back at the T^2 and SPE statistics of Module 2 from an entirely different direction.

Module 6: Virtual Metrology and Digital Twins

Metrology is the scarcest resource in a fab, and virtual metrology predicts a measurement from the sensor traces of the run that produced it: this module covers feature construction from raw traces, the $p \gg n$ problem, and the regularized and latent-variable models (lasso, PLS) that survive it. The second lecture then breaks the model it just built — exposing leakage from shuffled splits and from training on the future, establishing the metrology noise floor as the real accuracy ceiling, and working through drift, retraining triggers, per-chamber models, and where a virtual metrology model sits inside a digital twin.

Module 7: Test, Qualification, and Reliability

Test produces the largest labeled dataset in semiconductors with a positive class near 100 DPPM, and the first lecture follows that imbalance through the classical progression of PAT, DPAT, and spatial outlier screening to the escape–overkill trade-off curve and the cost function that sets the operating point — making the case that accuracy and F1 are the wrong metrics here. The second lecture turns to qualification, an extrapolation problem with almost no data, covering Arrhenius and Black acceleration models, censored lifetime distributions, and the Bayesian and physics-informed methods that earn their keep exactly when two models fitting the same data differ by a hundredfold in what they predict.

Module 8: Transfer Learning and Domain Adaptation

A fab never deploys one model — it maintains a fleet of them across chambers, tools, nodes, and products, each with too few labels to train from scratch. This module covers fine-tuning, feature reuse, domain adaptation, and hierarchical models for chamber matching and node-to-node transfer, and treats negative transfer as the central risk: it cannot be detected without target-domain labels you do not have.

Module 9: Generative Models for Scarce and Imbalanced Data

Generative models earn their place in a fab in one specific situation: when the data you need most is the data you can least afford to collect. Extending the progression from PCA through autoencoders to VAEs and diffusion models, this module covers physically-valid augmentation of rare defect classes, conditional generation for inverse design and computational lithography, and the evaluation problem — the only honest measure of a synthetic dataset is downstream performance on real held-out data.

Module 10: Physics-Informed Machine Learning

Every model so far in this course learned a function from data; a physics-informed model learns a function that must also satisfy a governing equation, which buys extrapolation and data efficiency rather than raw accuracy. This module places PINNs on a spectrum running from hard-coded physical laws through residual and hybrid models to pure black boxes, and makes the honest case for where they win — inverse problems, parameter identification, and fast operator surrogates — and where a finite element solver simply beats them.

Module 11: Bayesian Optimization and Design of Experiments

When each evaluation costs a wafer or an hour of simulation, the question is not how to fit a model but where to spend the next run. Building on Gaussian process regression from Module 6, this module develops acquisition functions, constrained and multi-objective optimization, and multi-fidelity methods that combine cheap simulation with expensive experiment — demonstrated on a thermal digital twin of a 3D logic-on-memory stack, where unconstrained optimization converges to a design that violates the memory temperature specification.

Module 12: Reinforcement Learning for Process and Design

Process control asks what to do now and optimization asks where to sample next; reinforcement learning asks what *sequence* of actions maximizes an outcome observed only at the end. This module introduces Markov decision processes, bandits, and policy learning, surveys where RL genuinely delivers in semiconductors — test-point insertion, place-and-route, adaptive test, autonomous experimentation — and confronts the sample-complexity problem that keeps RL inside simulators and design tools rather than on the fab floor.

Module 13: Agentic AI for Process Development

An LLM agent tunes an SF_6/O_2 plasma etch recipe by repeatedly calling a physics simulator, reading virtual metrology off the resulting profile, and revising — a closed loop rather than a chatbot demonstration, complemented by guest sessions on agentic AI for semiconductor data workflows and on LLM-based fab knowledge management. The module is the synthesis of the optimization sequence that precedes it, since the agent is performing optimization under uncertainty against a digital twin, and its performance is measured honestly against the search and Bayesian optimization baselines students have already built.

Module 14: AI in Design and Electronic Design Automation

Industry guest lecturers cover the design side of the semiconductor pipeline: design-technology co-optimization, data-driven compact modeling, computational lithography and the tapeout flow from design file to photomask, agentic AI in chip design, and simulation-driven design workshops. Together these close the loop between the design decisions that create a process requirement and the manufacturing and digital twin methods that the rest of the course addresses.

4. Assignments and Labs

4.1 Datasets

Every assignment in this course runs on a semiconductor dataset. Several are public collections used widely in the literature; the remainder are purpose-built and generated by the course notebooks from fixed seeds, with faults planted deliberately because the problems ask students to find them. All are distributed with the notebooks in the course repository, and each module notebook downloads its own data on first run.

Dataset	What it is	Modules
MIR-WM811K wafer maps	811,457 production wafer maps, about 21 percent labelled with one of nine failure patterns; severely imbalanced and organised by lot.	4, 8, 9
NFFA-EUROPE SEM dataset	18,577 annotated scanning electron microscope images in ten categories, released under a Creative Commons licence.	8
DAGM 2007	Ten sets of generated textured surfaces with one labelled defect per defective image; the standard weakly-supervised inspection benchmark.	8, 9
MVTec AD and AD 2	Fifteen object and texture categories for industrial anomaly detection, with pixel-level ground truth and defect-free training sets.	8, 9
SECOM	High-dimensional, heavily imbalanced fab sensor data with pass/fail labels, from the UCI repository.	2, projects
Tennessee Eastman process	The reference benchmark for process monitoring and multivariate fault detection.	2, projects
Process trace population	One hundred healthy runs and six injected fault types, generated by the course notebooks from fixed seeds.	2
Process control datasets	Critical dimensions as individuals and subgroups, an autocorrelated etch-rate series with seasoning drift, probe yield counts, a gauge study and a run-to-run loop.	5
Virtual metrology dataset	1,200 wafers across two chambers with 465 CD-SEM labels, raw traces, a maintenance log, and a measured 0.22 nm noise floor.	6
Wafer-sort parametric data	Parametric test vectors with die coordinates and lot-to-lot drift, together with censored lifetime data.	7
ViennaPS	Open-source level-set plasma etch simulator with Monte Carlo ray tracing; the process twin behind the agentic lab.	13
Stack thermal twin	Finite element thermal model of a 3D logic-on-memory stack, built on scikit-fem with an algebraic multigrid solver.	11

4.2 Software and Submission

Assignments are written as Python notebooks. Where you run them is your own choice — locally in VS Code or JupyterLab, or in Google Colab, for which every module notebook carries a link and downloads its own datasets. The environment is a suggestion; the repository is not.

Each student is required to maintain a personal GitHub repository for the semester, and all problem sets, the advanced lab and the project are submitted through it: one folder per assignment, with notebooks committed with their outputs saved so results can be read before anything is run. Version control, a readable commit history and a repository someone else can clone and run are the minimum conditions for the deployment practices covered in Module 3.

4.3 Module Problem Sets

Each application module ships a problem set as a Python notebook, distributed with its datasets through the course GitHub repository. The problems are worked against the same data the lectures use, so a result that contradicts a slide is a finding rather than an error.

Fault Detection and Classification. Generate the healthy run population, reproduce both failures from Lecture 1, and implement landmark alignment. Fit the multivariate SPC baseline, sweep the number of components and the control-limit percentile, and report the severity floor for each fault type.

Defect Classification. Reproduce the four wafer-map classifiers on MIR-WM811K under lot-grouped splits, and report macro-F1 with a class-by-class breakdown. Advanced options add directional features to recover the Scratch class, or pretrain on the 638,507 unlabelled wafers and fine-tune.

Statistical and Advanced Process Control. Compute z-scores and implied parametric yield, decompose variance across within-wafer, wafer-to-wafer and lot-to-lot terms, then build $\bar{X}$–R and I–MR charts and report Cp, Cpk, Pp and Ppk against spec. A third set adds a gauge R&R study that corrects the earlier capability number, residual charting on a drifting etch process, and a multivariate model charting T^2 and SPE with contribution plots.

Virtual Metrology. Build step-segmented features from the raw traces, beat the previous-measurement baseline with ridge, lasso and PLS, and establish the error floor implied by a 0.22 nm gauge σ. A second set re-evaluates that same model under lot-grouped, leave-one-chamber-out and time-ordered splits, detects the wet-clean regime change, and asks for a two-page deployment plan.

Test, Qualification, and Reliability. Build escape–overkill curves for two screens on wafer-sort parametric data, and write the cost function explicitly with justified escape and overkill costs. The deliverable is the cost analysis rather than the classifier: report how the optimal threshold moves when the cost ratio changes by a factor of ten in each direction.

4.4 Advanced Lab

Students complete one advanced lab in the second half of the semester. In the agentic process engineer lab, a language-model agent tunes an SF_6/O_2 plasma etch recipe by calling a level-set process simulator, reading virtual metrology off the resulting profile, and revising — evaluated honestly against random-search and Bayesian-optimisation baselines. In the design co-optimisation lab, students run constrained Bayesian optimisation against a finite element thermal twin of a 3D logic-on-memory stack and show that unconstrained expected improvement converges to a design that violates the memory temperature specification.

5. Semester Project

The semester project is the centerpiece of the course (80% of grade). Teams of 2–3 students will identify a semiconductor process or metrology problem, acquire or generate appropriate data, apply ML techniques studied in the course, and present their findings in both oral and written form.

5.1 Project Scope

Projects should address a concrete problem in semiconductor process technology, metrology, or manufacturing. Example topics include:

- ML surrogate model for a specific etch or deposition process using TCAD-generated or experimental data
- Defect classification system for a particular defect type using publicly available SEM or wafer-map datasets
- Virtual metrology model for predicting CD or overlay from equipment sensor logs
- Physics-informed neural network for modeling a thermal or diffusion process step
- Yield prediction model correlating inline metrology and process parameters to die-level yield
- Anomaly detection system for chamber health monitoring using synthetic or real sensor data
- Bayesian optimization framework for process recipe tuning in a simulated environment
- Equipment intelligence system: predictive maintenance, FDC, or digital twin for a specific process tool
- Agentic AI / MCP application for fab knowledge workflows or process recipe exploration
- Comparative study of ML architectures for a specific semiconductor data modality (time-series, images, tabular)

Students are encouraged to propose topics related to their own research. The instructor will help refine scope during office hours. All projects follow the CRISP-DM framework and ship as a GitHub repository containing a Jupyter walkthrough and a deployed Gradio demo.

5.3 Evaluation Criteria

Projects are evaluated on: (1) problem motivation and relevance to semiconductor technology, (2) appropriate selection and application of ML methods, (3) rigor of experimental methodology (proper data splits, baselines, ablations), (4) quality of analysis and interpretation of results, (5) clarity of written report and oral presentation, (6) novelty or insight contributed, (7) quantified manufacturing/business impact (yield, cost, cycle time, reliability), (8) discussion of corner cases and failure modes, and (9) quality of the deployed artifact: working GitHub repository with Jupyter walkthrough and Gradio demo.

6. Grading

Component	Weight
Semester Project	80%
Assignments, Quizzes and Short Tests	20%

Letter grades follow the Georgia Tech scale. The instructor reserves the right to curve final grades upward.

Late Policy: Assignments submitted up to 48 hours late will receive a 15% penalty. Submissions beyond 48 hours will not be accepted without prior arrangement. Project milestones cannot be submitted late.

7. Resources and References

There is no required textbook. The course draws on research papers, open-access tutorials, and instructor-prepared notes. The following references are recommended for supplementary reading:

7.1 Machine Learning Foundations

- A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.
- I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016. [Online]. Available: https://www.deeplearningbook.org
- C. M. Bishop and H. Bishop, *Deep Learning: Foundations and Concepts*. Cham, Switzerland: Springer, 2024.

7.2 Semiconductor Processes and Manufacturing

- G. S. May and C. J. Spanos, *Fundamentals of Semiconductor Manufacturing and Process Control*. Hoboken, NJ, USA: Wiley-IEEE Press, 2006.

8. Course Policies

8.1 Academic Integrity

All students are expected to comply with the Georgia Tech Academic Honor Code: http://www.policylibrary.gatech.edu/student-affairs/academic-honor-code. Collaboration on labs and homework is encouraged for discussion, but all submitted code and writing must be your own (or your team's, for the project). Use of AI coding assistants (e.g., GitHub Copilot, ChatGPT) is permitted for labs and homework as a learning tool, but must be disclosed. AI tools may not be used for paper critiques.

8.2 Accommodations for Students with Disabilities

If you have a disability and require accommodations, please contact the Office of Disability Services at https://disabilityservices.gatech.edu/ to establish your eligibility. Then provide the instructor with your accommodation letter as early as possible so that appropriate arrangements can be made.

8.3 Absences and Attendance

Attendance is expected for all lectures and is essential for industry guest lectures. The Institute Absence Policy applies: http://www.catalog.gatech.edu/rules/4/. Students who must miss class should notify the instructor in advance and are responsible for obtaining notes and materials from classmates.

8.4 Student-Faculty Expectations Agreement

At Georgia Tech we believe that it is important to continually strive for an atmosphere of mutual respect, acknowledgment, and responsibility between faculty members and the student body.

See http://www.catalog.gatech.edu/rules/22/ for an articulation of some combinated expectations that we have of each other and of ourselves.